A LaTeXML Demo

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Abstract

This document is a test to see how things look like when combining HTML files generated with LaTeXML and Jekyll posts. Other versions of this document are available:

- HTML.
- TeX.

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1 Math formulas

Unordered list

- Inline formula: $E = mc^2$.
- Another formula: The equation $ax^2 + bx + c = 0$ has at most two roots.

Ordered list

1. Element 1
2. Element 2

3. Element 3

Equation

$$E = mc^2 \tag{1.1}$$

$$\sqrt[3]{x^2 + 4x + 4} = ?$$

2 Theorems, Lemmas, etc

Theorem 2.1 (Basic Theorem)**.** *We always have ...*

Proof. It is well-known that ... □

Exercise 2.2. In this exercise, we apply Theorem 2.1 to ...

3 Algorithm

Here is an example of an algorithm.

Algorithm 1 An example algorithm

Input: The inputs are ...
Output: The outputs are ...

```
A statement
if The above statement is true then
    Do something
else
    Do nothing
end if
return something
```

Algorithm 1 computes ...

4 Cross references

As in Algorithm 1, we have ...

4.1 Test 1 of subsection

This is an illustration of a subsection. This is an illustration of a subsection. This is an illustration of a subsection. This is an illustration of a subsection. This is an illustration of a subsection.

4.2 Test 2 of subsection and paragraph

In this section, we will ... This is an illustration of a subsection. This is an illustration of a subsection. This is an illustration of a subsection.

Test paragraph This is an illustration of a paragraph. This is an illustration of a paragraph. This is an illustration of a paragraph.

5 Figure

A figure

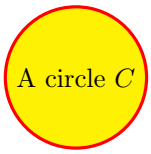

Figure 1: A figure

6 Tables

Number	Description
1	One
2	Two
3	Three

Table 1: A table.

multi-row	1	2	3
	4	5	6
multi-column			7

Table 2: A table with multi-row and multi-column.

GG&A Hoofed Stock			
Year	Price		Comments
	low	high	
1971	97–245		Bad year.
72	245–245		Light trading due to a heavy winter.
73	245–2001		No gnus was very good gnus this year.

Table 3: An example from LaTeXML Examples Page.

7 Source Code

Using `verbatim` environment.

```
\begin{table}[!ht]
\begin{tabularx}{\textwidth}{|X|X|}
\hline
Number & Description\\
\hline
1 & One \\
2 & Two \\
3 & Three \\
\hline
\end{tabularx}
\caption{A table.}
\end{table}
```

Using `listing` package.

```python
import numpy as np

def incmatrix(genl1,genl2):
    m = len(genl1)
    n = len(genl2)
    M = None #to become the incidence matrix
    VT = np.zeros((n*m,1), int)  #dummy variable

    #compute the bitwise xor matrix
    M1 = bitxormatrix(genl1)
    M2 = np.triu(bitxormatrix(genl2),1)

    for i in range(m-1):
        for j in range(i+1, m):
            [r,c] = np.where(M2 == M1[i,j])
            for k in range(len(r)):
                VT[(i)*n + r[k]] = 1;
                VT[(i)*n + c[k]] = 1;
                VT[(j)*n + r[k]] = 1;
                VT[(j)*n + c[k]] = 1;

                if M is None:
                    M = np.copy(VT)
                else:
                    M = np.concatenate((M, VT), 1)

                VT = np.zeros((n*m,1), int)

    return M
```

Listing 1: Python example

8 References and Citation

See [2], or [3], or [1].

References

[1] Jane Doe. Another paper. In *Proceedings of some great conferences*. Schloss Dagstuhl–Leibniz-Zentrum fuer Informatik, 2018.

[2] John Doe. An interesting article. *Journal of nice papers*, 10(1):1–20, 2018. doi: 10.1007/xxxxxxx.

[3] John Smith and Jane Doe. A nice preprint. *arXiv preprint*, 2018.